

國立高雄大學 113 學年度第 2 學期寒假轉學招生考試試題(轉二年級)

科目：微積分
考試時間：80 分鐘

系所：
電機工程學系(無組別)
本科原始成績：100 分

是否使用計算機：是

1. $\lim_{x \rightarrow 1} \left(\frac{x^9 - 1}{x^3 - 1} \right)$

2. $\frac{d}{dt} \left[\frac{1}{\sqrt{t^2 + 1}} \right]$

3. $\frac{d^2}{d\theta^2} [\sin(\theta^2)]$

4. $\sum_{k=2}^{\infty} \ln \left(1 - \frac{1}{k^2} \right)$

5. $\sum_{k=10}^{20} k^2$

6. $\int \left(\frac{1}{e^{2x} + 1} \right) dx$

7. $\int \left(\frac{1}{x^2 + 2x - 8} \right) dx$

8. $\int \left(\frac{1}{x^2 - 2x + 8} \right) dx$

9. $\frac{d}{d\theta} \left(\int_{-1}^{\tan \theta} \sqrt{1 + x^2} \, dx \right)$

10. $\int_0^2 |\sqrt{4 - x^2} - 1| \, dx$

國立高雄大學 113 學年度第 2 學期寒假轉學招生考試試題(轉二年級)

科目：物理

考試時間：80 分鐘

系所：

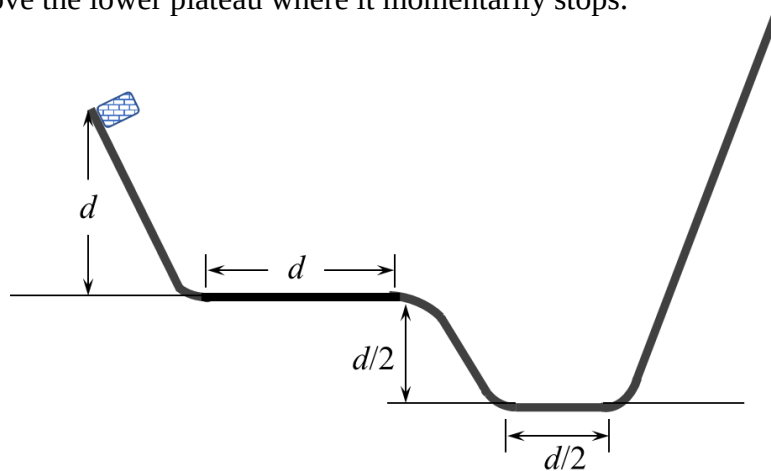
電機工程學系(無組別)

本科原始成績：100 分

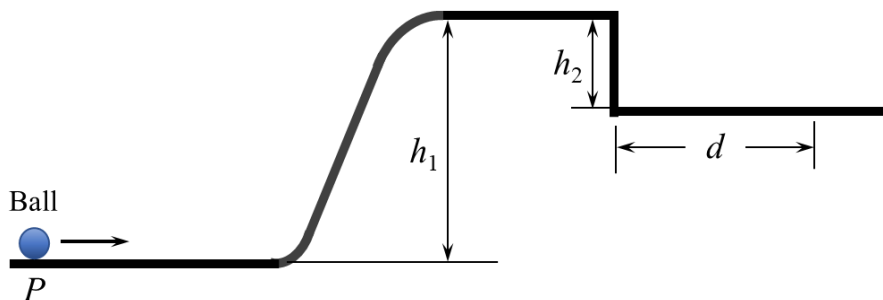
是否使用計算機：是

注意事項：計算題需有詳細計算過程，僅有答案將不予計分。

1. (12%) In the shown figure, a block is released from rest at height $d = 50$ cm and slides down a frictionless ramp and onto a first plateau, which has length d and where the coefficient of kinetic friction is 0.50. If the block is still moving, it then slides down a second frictionless ramp through height $d/2$ and onto a lower plateau, which has length $d/2$ and where the coefficient of kinetic friction is again 0.50. If the block is still moving, it then slides up a frictionless ramp until it (momentarily) stops. Where does the block stop? If its final stop is on a plateau, state which one and give the distance L from the left edge of that plateau. If the block reaches the ramp, give the height H above the lower plateau where it momentarily stops.



2. (12%) In the shown figure, a small, solid, uniform ball of mass M and radius R is to be shot from point P so that it rolls smoothly along a horizontal path, up along a ramp, and onto a plateau. Then, it leaves the plateau horizontally to land on a game board, at a horizontal distance d from the right edge of the plateau. The vertical heights are $h_1 = 5.00$ cm and $h_2 = 1.60$ cm. With what speed must the ball be shot at point P for it to land at $d = 6.00$ cm? (The moment of inertia of a solid sphere about central axis is $(2/5)MR^2$.)



3. (5%) Explain why the surface of a perfect conductor is an equipotential surface under electrostatic condition.

科目：物理

考試時間：80 分鐘

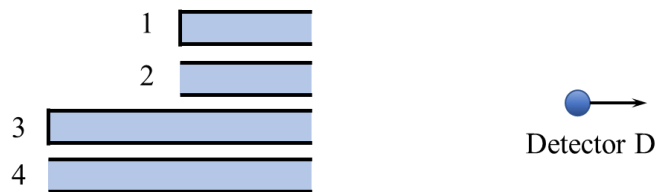
系所：

電機工程學系(無組別)

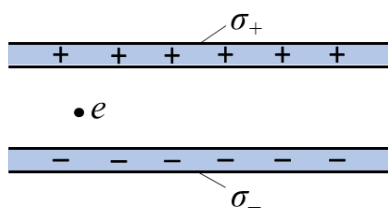
本科原始成績：100 分

是否使用計算機：是

4. (12%) The figure shows four tubes with lengths 1.0 m or 2.0 m, with one or two open ends as drawn. The third harmonic is set up in each tube, and some of the sound that escapes from them is detected by detector D , which moves directly away from the tubes. In terms of the speed of sound v , what speed must the detector have such that the detected frequency of the sound from (a) tube 1, (b) tube 2, (c) tube 3, and (d) tube 4 is equal to the tube's fundamental frequency?

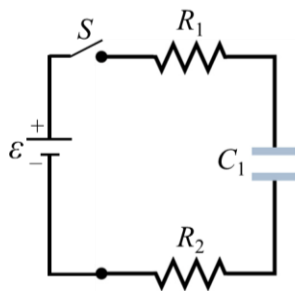


5. (10%) In the shown figure an electron is released between two infinite nonconducting sheets that are horizontal and have uniform surface charge density σ_+ and σ_- , as indicated. The electron is subjected to the following three situations involving surface charge densities and sheet separations. Rank the magnitudes of the electron's acceleration and Explain your answer. (無說明理由將不予計分)

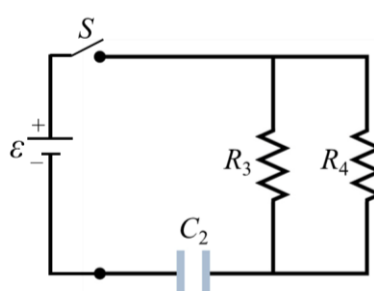


Situation	σ_+	σ_-	Separation
1	$+4\sigma$	-4σ	d
2	$+7\sigma$	$-\sigma$	$4d$
3	$+3\sigma$	-5σ	$9d$

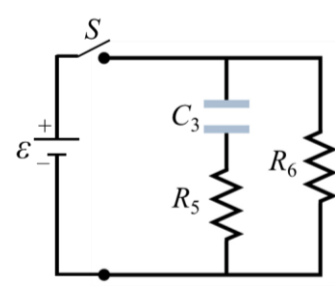
6. (15%) Consider three circuits as shown and assume the batteries have no internal resistance. All capacitors (C_1 , C_2 , and C_3) are initially uncharged. The resistors and capacitors are all identical with resistance R and capacitance C , respectively. (a) All switches S are closed at $t = 0$, find the currents in the resistors R_2 , R_4 , and R_6 at $t = 0$? After the switches are closed for a very long time ($t \rightarrow \infty$), (b) determine the currents in the resistors R_2 , R_4 , and R_6 and (c) the stored energies in the capacitors C_1 , C_2 , and C_3 . (d) Rank the time required for the capacitor to reach 50% of its final charge and Explain your answer. (無說明理由將不予計分)



(1)



(2)



(3)

科目：物理

考試時間：80 分鐘

系所：

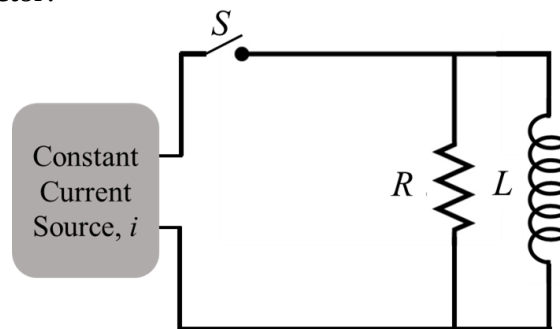
電機工程學系(無組別)

本科原始成績：100 分

是否使用計算機：是

7. (11%) An electron moves in a circle of radius $r = 5.29 \times 10^{-11} \text{ m}$ with speed $2.19 \times 10^6 \text{ m/s}$. Treat the circular path as a current loop with a constant current equal to the ratio of the electron's charge magnitude to the period of the motion. If the circle lies in a uniform magnetic field of magnitude $B = 7.10 \text{ mT}$, what is the maximum possible magnitude of the torque produced on the loop by the field?

8. (12%) In the shown figure, after switch S is closed at time $t = 0$, the emf of the source is automatically adjusted to maintain a constant current i through S . (a) Find the current through the inductor as a function of time. (b) At what time is the current through the resistor equal to the current through the inductor?



9. (11%) Two light rays go through different paths by reflecting from the various flat surfaces as shown. The light waves have a wavelength of 420.0 nm and are initially in phase. What are the (a) smallest and (b) second smallest value of distance L that will put the waves exactly out of phase as they emerge from the region?

